Juan Zurita

Teaching Statement

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I teach economics as a quantitative, computational discipline. My courses—from large introductory lectures to specialist Honours seminars—share a common design principle: students learn economic ideas best when they can build, estimate, and break the models themselves. Over fifteen years across Argentina, Australia, and the United Kingdom, I have taught macroeconomics, econometrics, forecasting, and programming to students ranging from first-year undergraduates to PhD candidates. At the University of Edinburgh, I designed and deliver *Programming and Numerical Methods for Economics*, a course that equips economics students with the computational and quantitative tools their research and careers increasingly demand.

Teaching Philosophy

Three core principles guide my course design and delivery:

Theory and computation together, not in sequence. Economics students too often encounter mathematical models in lectures and empirical data in separate lab sessions, with little connection between the two. I structure my courses so that computation is the medium through which students engage with theory. In *Programming and Numerical Methods*, for instance, students implement a Cobb-Douglas production function, compute its marginal products numerically and via automatic differentiation (JAX), and compare both to the analytical solution—all in a single session. The exercise teaches Python syntax while transforming abstract concepts like diminishing returns into observable, testable phenomena.

Concrete before abstract. I introduce specific economic questions or real-world datasets before diving into formal modeling. In *Economics 2* at Edinburgh, I open the fiscal policy module not with an abstract IS-LM derivation, but with an empirical question students lived through: "Did the UK furlough scheme work?" The formal framework then enters as a functional tool to evaluate the answer, giving students a intuitive mental anchor before engaging with the underlying algebra.

Active problem-solving in real time. To prevent passive learning, I regularly interrupt lectures with short, targeted diagnostic exercises—such as predicting code output or identifying key parameter drivers. In larger introductory cohorts, I utilize live interactive polling (e.g., Wooclap) to identify and address student misconceptions in real time, making confusion visible early when it is easiest to correct.

Course Design and Innovation

Programming and Numerical Methods for Economics (Honours, Edinburgh, 2024–present). I designed this course to equip students with computational tools to solve, simulate, and estimate the quantitative models used in advanced research and Honours dissertations. The syllabus covers Python fundamentals, data manipulation with Pandas, numerical optimization, root-finding, interpolation, automatic differentiation with JAX, and Monte Carlo methods. Students apply these techniques directly to economic problems: interpolating value functions in consumption-savings decisions, estimating impulse responses from VARs, and computing numerical elasticities. All lecture materials, interactive Jupyter notebooks, and problem sets are publicly accessible via my course website.

Proposed Course: Deep Learning for Macroeconomics. Building on my computational curriculum, I have developed a course proposal for an advanced seminar introducing neural-network methods for solving and estimating macroeconomic models. Designed around interactive notebooks, the syllabus covers feed-forward and recurrent architectures, dynamic programming approximations, and likelihood-free inference, preparing students at the intersection of structural macroeconomics and machine learning.

Core Teaching Experience

Macroeconomics (Undergraduate and Postgraduate). I have taught macroeconomics across all levels of university instruction. At Edinburgh, I lecture *Economics 2* (core macroeconomics, 2024–2026), covering national accounting, IS-LM, monetary and fiscal policy, and open-economy dynamics. At the University of Technology Sydney (UTS), I taught *Intermediate Macroeconomics*, *Advanced Macroeconomics* (Honours), and *Macroeconomics 2* (PhD level), covering dynamic programming, incomplete markets, and quantitative solution methods. At the Catholic University of Argentina (UCA, 2011–2015), I taught *History of Economic Thought*, tracing macroeconomic theory from classical foundations through Keynesian synthesis to modern DSGE modeling.

Econometrics and Forecasting. At the Australian National University (ANU, 2023), I taught *Economic and Business Forecasting*, covering time-series analysis, VAR estimation, and empirical forecast evaluation. At Edinburgh (2026), I teach *Applications of Econometrics* (Honours), where students utilize regression, instrumental variables, and panel-data methods to replicate published empirical literature.

Central Banking and Financial Policy. At UTS (2024), I taught *Central Banking and Monetary Policy*, drawing directly on my research in monetary transmission and bank balance sheets. Students analyzed central bank communications, estimated Taylor rules, and evaluated the distributional implications of quantitative easing.

Supervision

I actively supervise Honours and Master's dissertations at Edinburgh, guiding students through research design, data collection, empirical identification, and structural estimation. Supervised student projects include the macroeconomic impact of housing-market shocks, monetary policy pass-through to mortgage rates, and machine-learning applications in inflation forecasting.

Teaching Across Cultural and Institutional Contexts

Teaching across Argentina, Australia, and Scotland has provided valuable experience with diverse student backgrounds and academic traditions. At UCA, I taught in Spanish to cohorts receiving broad social science training; at UTS, I instructed a heavily international student body focused on applied, industry-ready skills; at Edinburgh, I teach high-performing students in a research-intensive environment. Adapting across these environments has reinforced that rigorous structure, clear exposition, and active engagement resonate universally across diverse student demographics.

Contributions to Departmental Teaching

I am prepared to contribute to the department's teaching mission in three ways:

1. **Core Curriculum Coverage:** Delivering high-quality lectures in core undergraduate and graduate macroeconomics, econometrics, and monetary economics.
2. **Computational & Deep Learning Electives:** Offering specialist coursework in Python, numerical methods, and machine learning applied to macroeconomics.
3. **Student Research Pipeline:** Supervising empirical and quantitative dissertations at the intersection of macroeconomics, banking, and financial policy.

Teaching Evaluations

Full evaluation reports available at my teaching page.

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